Part-II

Course No.	Title of the course	Credits	Marks	Unit
Chem 210F	Physical Chemistry II (Concepts of Physical Chemistry)	4.0	100	1
Chem 220F	Inorganic Chemistry II (Chemistry of Main Group Elements)	4.0	100	1
Chem 230F	Organic Chemistry II (Fundamentals of Organic Chemistry II)	4.0	100	1
Chem 232H	Fundamental of Pharmaceutical Chemistry	2.0	50	0.5
Chem 245F	Nuclear and Radiochemistry	4.0	100	1
Chem 250F	Industrial Chemistry	4.0	100	1
Math C251H	Mathematical Methods	2.0	50	0.5
Phys C261H	Electronics and Optics	2.0	50	0.5
Chem 215LH	Physical Chemistry Practical II	2.0	50	0.5
Chem 225LH	Inorganic Preparation and Volumetric Analysis	2.0	50	0.5
Chem 235LH	Preparative Organic Chemistry	2.0	50	0.5
Chem 280VH	Viva-Voce	2.0	50	0.5
	Total	34.0	850	8.5

Part-III

Course No.	Title of the course	Credits	Marks	Unit
Chem 310F	Physical Chemistry III (Selected Topics in Physical Chemistry)	4.0	100	1
Chem 312H	Polymer Chemistry	2.0	50	0.5
Chem 320F	Inorganic Chemistry III (Advanced Concepts)	4.0	100	1
Chem 322H	Inorganic Chemistry IV (Selected Topics)	2.0	50	0.5
Chem 324H	Molecular Symmetry and Group Theory	2.0	50	0.5
Chem 330F	Organic Chemistry III (Advanced Organic Chemistry)	4.0	100	1
Chem 332F	Chemistry of Natural Products	4.0	100	1
Chem 334H	Advanced Stereochemistry	2.0	50	0.5
Chem 340F	Chemical Spectroscopy	4.0	100	1
Chem 315LF	Physical Chemistry Practical III	4.0	100	1
Chem 335LF	Identification of Organic Compounds	4.0	100	1
Chem 370LH	Computing and Information Technology for Chemistry	2.0	50	0.5
Chem 380VH	Viva-Voce	2.0	50	0.5
	Total	40.0	1000	10.0

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Course Code: Chem. 310F	Course Title: Physical Chemistry III (Selected Topics in Physical Chemistry)	
Year: 3rd	Credit: 4	Unit: 1
Marks: 10+20+70 = 100	Contact Hours: 55-60	Session: 2021-2022, 2022-2023
Type: Core Theory	Course Teacher(s): Will be assigned before course commencement.	

Rationale

This course provides the knowledge on electric and magnetic proportion of substances, fundamentals of photochemistry, reaction catalysis, adsorption isotherms, colloidal systems as well as phase diagram of two and three component system.

Course Objectives

The learning objectives of this course are to

- extend student's knowledge on electric and magnetic properties of substances.
- understand fundamentals of photochemical reactions and catalysis.
- impart knowledge on surface chemistry, colloids and phase diagrams.

Course Content	Learning Outcomes
1. Electric properties of molecules: Polar and non-polar molecules; bond moment and dipole moment; polarization, polarizability; Clausius-Mossotti equation; Debye equation; dependence of polarizability on frequency; measurement of dipole moment: Ebert's method, vapour-temperature method, refraction method; application of dipole moment in elucidation of ionic character, bond moment and geometry of molecules, resonance structure, determination of bond length. 2. Magnetic properties of substances: Magnetic permeability and susceptibility; paramagnetic, diamagnetic and ferromagnetic substances; theories of paramagnetism; theories of diamagnetism; measurement of magnetic susceptibility: Gouy method, Bhatnagar-Mathur method, Quincke's method, Curie's balance; application of magnetic susceptibility measurements. 3. Photochemistry: Dark and photochemical reactions; laws of photochemistry; consequences of light absorption: primary and secondary processes; potential energy curves for primary photochemical processes; quantum efficiency; determination of quantum yield; factor affecting quantum efficiency, reasons for low and high quantum yield; fluorescence; phosphorescence; chemiluminescence; application of fluorescence and phosphorescence; photosensitization; mechanism and kinetics of photochemical reactions, photochemical chain reactions: hydrogen-chlorine and hydrogen-bromine reactions; non-chain photochemical reactions: dissociation of hydrogen iodide, photolysis of acetaldehyde; lasers and its applications. 4. Catalysis: Types and characteristics of catalysis; theory and kinetics of homogeneous catalysis; function of a catalyst in terms of Gibbs free energy of activation; theory and kinetics of heterogeneous catalysis; effect of temperature on heterogeneous reactions; acid-base catalysis; mechanism and kinetics of acid-base catalysis. 5. Surface chemistry: Adsorption, absorption and sorption; physisorption and chemisorption; adsorption isotherms: Freundlich, Langmuir's and BET isotherms; measurement of surface area of adsorbent; heat of adsorption; Gibbs adsorption equation; surface excess; applications of adsorption; surface films. 6. Colloids: Colloidal dispersions; classification of colloids; lyophilic and lyophobic colloids; preparation and properties of sols; optical properties: Tyndall effect; Brownian movement; determination of Avogadro's number;	On successful completion of this course, students should be able to • describe permanent dipoles and induced dipoles, measure dipole moments and apply dipole moments in elucidation of molecular geometry. • identify paramagnetic, diamagnetic and ferromagnetic substances, measure magnetic susceptibility and apply the data for useful purposes. • describe the photochemical laws, primary and secondary photochemical processes, determine quantum yield and interpret the data, describe photochemical chain and non-chain reactions. • discuss the theory of homogeneous and heterogeneous catalysis, acid-base catalysis. • classify colloids, explain optical and electrical properties of colloids, emulsions, gels and micelles. • explain two component and there component phase diagrams.

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stability of sols; electrical properties: electrophoresis, electro-osmosis, streaming and sedimentation potential; origin of charge, electrical double layer, zeta potential; emulsions; gels; micelles; applications of colloids. 7. Phase diagram: Phases, components, and degrees of freedom; Phase rule; two-component systems: vapour pressure diagrams, isopleths, tie line, lever rule, temperature-composition diagrams, distillation of mixtures, azeotropes, liquid-liquid, liquid-solid phase diagrams, phase separation, critical solution temperature, partially miscible liquids, role of added salts; distillation of partially miscible liquids, eutectics, incongruent melting; three-component systems: triangular phase diagrams.	

Assessment and Evaluation: Attendance 10%, 3 tutorials 20% and year end final exam 70%.

Suggested Readings

1. Physical Chemistry, Peter W. Atkins, and Julio de Paula, W. H. Freeman and Co.
2. Advanced Physical Chemistry, Gurdeep Raj, GOEL Publishing House, Meerut, India.
3. Elements of Physical Chemistry, S. Glasstone and D. Lewis, Macmillan Education.
4. Advanced Physical Chemistry, D. N. Bajpai, S. Chand Co. Ltd, New Delhi, India.
5. Modern Photochemistry, N. J. Turro.
6. Surface Chemistry, Reigel, Arft and Jacob.

Course Code: Chem. 312H	**Course Title:** Polymer Chemistry	
Year: 3rd	**Credit:** 2	**Unit:** 0.5
Marks: 5+10+35 = 50	**Contact Hours:** 35-40	**Session:** 2021-2022, 202
Type: Core Theory	**Course Teacher(s):** Will be assigned before course commencement.	

Rationale

This course is designed to offer a general idea of polymers including preparation, classification and its physical, and mechanical properties.

Course Objectives

The learning objective of this course is to

- Provide a general idea of polymeric compounds, their molecular weight distribution, properties in preparation methods and mechanical properties.

Course Content	Learning Outcom
1. Polymer Structure: Definition of polymer, difference between polymers and macromolecule; classification of polymers; degree of polymerization, nomenclature and tacticity; basic structure of polymers (linear and branched polymers; moderately cross linked polymer); molecular forces and chemical bonding in polymers. 2. Polymer solutions: Criteria for polymer solubility; size and shapes of polymers in solution; thermodynamics of polymer solution; Flory-Huggins theory, fractionation of polymers by solubility. 3. Molecular weight and size of polymers: Number average, molecular weight average, Z-average and viscosity, average molecular weight; distribution of molecular weight; determination of molecular weight by end group analysis, osmotic pressure measurement, light scattering, viscosity measurement and ultra-centrifugation. 4. Polymerization: Types of polymerization; (a) step-reaction (condensation) polymerization: mechanism and kinetics of stepwise polymerization, statistics and molecular weight control; (b) Radical chain (addition)	Upon completion of this the student will be ab • describe the definition, classification, configur and conformation, nomenclature of polyn • explain the molecular and distribution of mo weight • measure molecular we different methods • describe the solubility, shape and thermodyna polymer solutions • understand the differe polymerization proces kinetics and thermody

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polymerization: mechanism-initiation, propagation, termination, kinetics and thermodynamics of radical polymerization, degree of polymerization and chain transfer, ceiling temperature; (c) Ionic polymerization: Similarities and contrasts in ionic polymerization, mechanism and kinetics of cationic and anionic polymerization, living polymers. 5. Preparation with mechanism, properties and uses: Polyethylene, polystyrene, phenol-formaldehyde resins, melamine-formaldehyde resins, urea-formaldehyde resins, epoxy resins, polyester, PVC, polyamide. 6. Rheology and mechanical properties of polymers: Flow behavior of polymers: Newtonian and non-Newtonian flow; elasticity, viscoelasticity (viscoelastic properties of polymers); The glass transition temperature (Tg); the mechanical properties of crystalline polymers: crystallinity, the crystalline melting point etc.	• discuss the preparation, properties and uses of some important polymers • explain the flow behaviour and mechanical properties of polymers. • analyze values of different properties of polymers. • develop concept for preparing new polymers.

Assessment and Evaluation: Attendance 10%, 3 tutorials 20% and year end final exam 70%.

Suggested Readings

1. Textbook of Polymer Science, Padmal L. Nayak, S. Lenka.
2. Polymer Science, V. R. Gowarker, N. V. Viswanathan and Jayadev Sreadhar.
3. Introduction to Polymer Chemistry (International Student Editions), R. B. Seymour.
4. Polymer Chemistry, M. G. Arora, M. Singh.
5. Text Book of Polymer Science (Willey), Fred W. Billmeyer.
6. Introductory Polymer Chemistry, G. S. Misra, Wiley Eastern Limited, India.

Course Code: Chem. 320F	**Course Title:** Inorganic Chemistry III (Advanced Concepts)	
Year: 3rd	**Credit:** 4	**Unit:** 1
Marks: 10+20+70 = 100	**Contact Hours:** 55-60	**Session:** 2021-2022, 2022-2023
Type: Core Theory	**Course Teacher(s):** Will be assigned before course commencement.	

Rationale

This course is designed to provide advanced concepts of chemical bonding, molecular orbital theories, crystal field theories and their application, inorganic reaction mechanism as well as the magnetic properties of the coordination complexes.

Course Objectives

The learning objectives of this course is to

- convey knowledge on quantum mechanical interpretation of atomic structure and chemical bonding based on Schrödinger wave equation.
- provide knowledge about valence bond and molecular orbital theories of covalent bonding in the light of wave equation.
- promote knowledge about Pauling's valence bond, crystal field and molecular orbital theories of coordination compounds.
- impart knowledge on structural, thermodynamic and magnetic properties of coordination compounds.
- convey knowledge on the inorganic reaction mechanism of coordination complexes and redox reactions.
- impart knowledge on magnetic properties of molecules and different techniques on the measurement of magnetic susceptibility.

Course Content	Learning Outcomes
1. Quantum mechanical approach of atomic structure: Nature of wave motion, the Schrödinger wave equation, interpretation of the wave function, solution of the wave equation, Schrödinger equation for hydrogen like atom, separation of variables, the radial and angular functions and probability functions, representation of symmetry orbitals.	Upon completion of this course the students will be able to • understand and apply the concepts of bonding theories (e.g. valence bond and

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2. Detailed study of the covalent bond theories: (a) The nature of the co-valent bonding: wave mechanical principle, valence bond theory (VBT) of hydrogen molecule, hybridization, VBT description using hybrid orbitals, resonance. (b) Molecular orbital theory (MOT) of homo- and hetero-diatomic molecules, HOMO and LUMO, molecular orbitals of polyatomic molecules with σ- and π-bonding, the ligand group orbital approach, comparison of VBT and MOT. 3. The chemistry of transition metals and co-ordination compounds: (a) The transition elements: Electronic configuration, special properties, (b) Elements of coordination chemistry: Nomenclature for coordination compounds, Werner's theory, electronic interpretation of co-ordination compounds, Pauling's valence bond theory, isomerism stereochemistry of 4-and 6-coordination complexes, the chelate complexes and examples of chelates, stabilization of unusual oxidation states by complexation applications of coordination compounds. 4. Advanced treatment of bonding in co-ordination compounds: The crystal field theory and the ligand field theory for octahedral and tetrahedral complexes, tetragonally distorted octahedral complexes and square planer complexes, molecular orbital theory: metal ligand σ-bonding and effect of π-bonding, ligand field stabilization energy, consequence of ligand field splitting: magnetic properties, electronic properties, spectrochemical series, structural and thermodynamic effects of ligand field splitting, the Jahn Teller effect. 5. Reaction mechanism of d-block complexes: (a) Ligand substitution reactions. Substitution in square-planar complexes, the *trans* effect, substitution in octahedral complexes, (b) redox reactions, (c) photochemical reactions. 6. Origin of magnetism, Ferromagnetism, Ferrimagnetism, Anti-ferromagnetism, Diamagnetism, Pascle's correction, Curie's Law, Curie-Weiss law. Molecular magnetism, spin angular moment, orbital angular moment, Bohr magneton, spin-orbital coupling, calculation of room temperature magnetic moment, variable temperature magnetic moment, SQUID, spin only magnetic moment, Van vleck's equation for magnetic susceptibility, intra- and intermolecular ferromagnetism, super exchange, variation of susceptibility with temperature, mathematical formulation for spin exchange in binuclear complexes.	molecular orbital theories) a to use these concepts in problem solving matters. • explain the structural featu of coordination compound • apply CFT (crystal field th to octahedral, tetrahedral a square planar complexes to explain the electronic, thermodynamic and magne properties of complex compounds. • understand the mechanism ligand substitution reaction square planar and octahedr complexes. • understand the magnetic properties of the molecule

Assessment and Evaluation: Attendance 10%, 3 tutorials 20% and year end final exam 70%.

Suggested Readings

1. Inorganic Chemistry-Principles of Structure and Reactivity, James E. Huhee, E. A. Keiter and R. L. Ke Ed., Pearson education.
2. Inorganic Chemistry, 3rd Ed., C. E. Housecroft and A. G. Sharpe, Pearson, USA.
3. Inorganic Chemistry, 3rd Ed., G. L. Miessler and D. A. Tarr, Pearson, India.
4. Inorganic Chemistry, D. F. Shriver and P. W. Atkin, Oxford University Press, UK.
5. Molecular Magnetism, Olivier Kahn, VCH Publishers, NY, USA.

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Course Code: Chem. 322H	**Course Title:** Inorganic Chemistry IV (Selected Topics)	
Year: 3rd	**Credit:** 2	**Unit:** 0.5
Marks: 5+10+35 = 50	**Contact Hours:** 35-40	**Session:** 2021-2022, 2022-2023
Type: Core Theory	**Course Teacher(s):** Will be assigned before course commencement.	

Rationale

The contents of the course is designed to provide through knowledge about the chemical and biological functions of metals/atoms in living beings.

Course Objectives

The learning objectives of this course are to

- impart knowledge on some aspects of inorganic chemistry, bonding in metals and alloys, non-aqueous solvent system, electron deficient and nonstoichiometric compounds, the polymer chemistry of carbon silicon, boron, phosphorus and nitrogen.
- promote knowledge on bioinorganic chemistry, Lanthanides and actinide chemistry.

Course Content	Learning Outcomes
1. Bonding in metals and alloys: The theory of metals, valence bond approach, the band theory of metals, metallic properties, solid solutions, supper structure and intermetallic compounds. 2. Non-aqueous solvent system: Classification of solvents, properties of ionizing solvents, solubility criteria in ionizing solvents, acid base phenomenon in non-aqueous system. Studies of some typical non-aqueous ionizing solvents such as liquid ammonia, liquid sulphur dioxide, liquid hydrogen sulphide and anhydrous sulphuric acid. 3. Electron deficient and nonstoichiometric compounds: (a) Boron hydrides and their derivatives, properties and studies of their bonding and structures; the borohydrides, carbaborane clusters, possible isomers of carbaboranes, applying Wade's rules to carbaborane structures, (b) Metal carbides; classification, structure, formation and uses. 4. Inorganic polymers: Chain and network polymers, silicon polymers; boron nitrogen polymers; sulphur-nitrogen polymers; phosphonitrillic compounds; fluorocarbons: preparation, structure and uses; chemistry of fullerenes. 5. Bioinorganic chemistry: Metalloporphyrins: chlorophyll, cytochrome, hemoglobin and myoglobin, vitamin B and B_{12} coenzymes; iron-sulphur proteins, nitrogen fixation. 6. Lanthanoids and Actinoids: Occurrence, electronic configuration, position in periodic table, general chemistry, separation techniques, Lanthanide compounds as NMR shift reagents; actinides: occurrence, electronic configurations, oxidation states, general chemistry, organometallic chemistry of actinides.	Upon completion of this course the students will be able to • describe selected topics of inorganic chemistry as well as some basic idea about NMR shift reagent. • understand the bonding and properties of metals. • know the nature and properties of non-aqueous solvents and bonding in electron deficient compounds. • explain inorganic polymers and the role of metals in living systems. • study metal containing biomolecules and their structures.

Assessment and Evaluation: Attendance 10%, 3 tutorials 20% and year end final exam 70%.

Suggested Readings

1. Advanced Inorganic Chemistry, F. A. Cotton and G. Wilkinson, John Willey and Sons, Inc. New Delhi.
2. Inorganic Chemistry-Principles of Structure and Reactivity, James E. Huhee, E. A. Keiter and R. L. Keiter, 4th Ed., Pearson education.
3. Non-aqueous solvents (Study in Modern Chemistry), T. C. Waddington, Thomas Nelson and Sons. Ltd.
2. An Introduction to Inorganic Chemistry, K. F. Purcell and J. C. Kotz, Saunders College Publishing.
5. Inorganic Chemistry, D. F. Shriver and P. W. Atkins, Oxford University press, UK.
6. Bioinorganic Chemistry, I. Bertini, H. B. Gray, S. J. Lippard, J. S. Valentine, University Science Books, CA, USA.

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Course Code: Chem. 324H	**Course Title:** Molecular Symmetry and Group Theory	
Year: 3rd	**Credit:** 2	**Unit:** 0.5
Marks: 5+10+35 = 50	**Contact Hours:** 35-40	**Session:** 2021-2022, 2022-2023
Type: Core Theory	**Course Teacher(s):** Will be assigned before course commencement.	

Rationale

This course is designed to provide the fundamental concepts of molecular symmetry and its application in the study chemical bonding and vibrational spectroscopy.

Course Objectives

The learning objectives of this course are to

- give the students the idea about the symmetry elements and operations.
- know the applications of symmetry in chemical bonding.
- know the application of symmetry in vibrational spectroscopy.

Course Content	Learning Outcomes
1. Symmetry elements and operations: Definition of the five symmetry elements (identity, proper rotation axis of order n, plane of symmetry, inversion center and improper axis of order n) and associated symmetry operations. 2. Point groups: Point group symbol, systematic classification of molecules into point groups, complete set of symmetry operations of a molecule constitutes a group. 3. Non-degenerate representations: Forming a non-degenerate representation to describe the effect of the symmetry operations of a group on a direction such as x, reducing a reducible representation to its component irreducible representations. 4. Matrices: Combining two matrices, setting up a matrix to perform a given transformation, finding the character of a matrix representing a symmetry operation using any given basis. 5. Degenerate representations: Finding the characters of a set of representations generated by using a set of degenerate vectors as a basis. 6. Applications to chemical bonding: Finding sets of hybrid orbitals with given directional properties, determining orbitals suitable for π-bonding in a molecule, finding the symmetries of LCAO molecular orbitals, constructing simple MO correlation diagrams. 7. Applications to molecular vibration: Finding the symmetry species of the normal modes of vibration of a molecule of a given symmetry, finding the number of infrared and Raman active vibrations in a molecule, finding the number of active vibrations in a characteristic region of the infrared or Raman spectrum of a molecule. 8. Linear combinations: Finding the combinations of bond stretching vibrations which form the bond stretching vibrational modes of a molecule, finding the symmetry adapted linear combinations of orbitals suitable for combining with the atomic orbitals of a central atom to form molecular orbitals, finding the form of the wave functions of hybrid orbitals, normalizing any of the above functions, confirming the orthogonality of normalized functions.	Upon completion of this cour the students will be able to • understand the symmetry elements, operation and point group of simple molecules. • understand the applications symmetry in chemical bond and vibrational spectroscopy.

Assessment and Evaluation: Attendance 10%, 3 tutorials 20% and year end final exam 70%.

Suggested Readings

1. Molecular Symmetry and Group Theory, Alan Vincent, John Wiley and Sons Ltd.
2. Basic Inorganic Chemistry, F. A. Cotton, G. Wilkinson and P. Gauss, Wiley, John Wiley and Sons.
3. Concepts and Models of Inorganic Chemistry, B. E. Douglas, D.H. McDaniel, J. J. Alexander, 3rd ed, John Wi Sons, Ltd. NY, USA.
4. Inorganic Chemistry, D. F. Shriver and P. W. Atkins, Oxford University Press.

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Course Code: Chem. 330F	Course Title: Organic Chemistry III (Advanced Organic Chemistry)	
Year: 3rd	Credit: 4	Unit: 1
Marks: 10+20+70 = 100	Contact Hours: 55-60	Session: 2021-2022, 2022-2023
Type: Core Theory	Course Teacher(s): Will be assigned before course commencement.	

Rationale

This course has been offered to develop the capacity for analyzing the different spectral data of simple organic compounds and the basic concept of stereochemistry. The course also provide advance knowledge about different classes of organic reactions and their mechanisms.

Course Objectives

Learning objectives of this course are to

- impart knowledge on analysis of IR, MS, UV, and NMR spectral data to elucidate structures of simple organic compounds.
- provide in-depth knowledge of the *cis-trans* isomerism and their interconversion, conformational analysis, relative and absolute configuration and their application.
- understand the role of identification of products and intermediates in the elucidation of reaction mechanisms.
- lay a rock-solid foundation in understanding of all major types of organic reaction mechanisms viz., substitution, elimination, addition.

Course Content	Learning Outcomes
1. Spectroscopy in analysis of organic compounds: A brief introduction and applications of IR, UV, NMR and Mass spectrometry in structural elucidation of simple organic compounds. 2. Organic reaction mechanisms: Classes of reaction mechanism; methods for determining reaction mechanisms: isolation of products, trapping and detection of intermediates, isotope tracing, kinetics and stereochemical evidence. 3. Broad concept of the mechanism of the following classes of organic reactions: (a) Nucleophilic substitution reactions in aliphatic system: S_N1 and S_N2, S_Ni reactions-kinetics, effect of solvent, effect of structure, effect of attacking reagents and leaving groups, stereochemistry, neighboring group participation, steric effect; allylic rearrangement. Aromatic nucleophilic substitution reactions; (b) Elimination reactions: E1, E2 and E1cB mechanisms; orientation in elimination reactions; competition between elimination and substitution; intramolecular (cis) elimination; (c) Addition reactions: Addition to carbon-carbon multiple bonds: addition of halogens; addition of hydrogen halides; peroxide initiated addition of hydrogen bromide; hydration of olefins. Mechanism and stereochemistry of nucleophilic additions, Diel's -Alder reactions, hydroxylation-cis and trans. 4. Name reactions: Wolf-Kishner reduction, Clemmensen reduction, Meerwein-Ponndorf-Verley reduction, Oppenauer oxidation, Baeyer-Villiger oxidation, reduction with metal hydrides, $LiAlH_4$, $NaBH_4$. 5. Heterocyclic compounds: Chemistry and structure of five and six-membered heterocyclic compounds containing two hetero atoms: imidazole, pyrimidine, oxazole and thiazole. Heterocyclic compounds having fused rings: chemistry and structure of indole, benzofuran, benzothiophene, quinoline and isoquinoline.	Upon completion of this course students will be able to • analyse IR, MS, UV, and NMR spectral data to elucidate structures of simple organic compounds. • acquire the skills for correct conformational and configurational assignment and understand cis-trans isomerisms. • describe the methods of determining reaction mechanism and explain the role of identification of products and intermediates in the elucidation of mechanisms. • write a detailed mechanism with supporting evidence of substitution, elimination and addition reaction. • propose rational and logical mechanism(s) for any organic reaction run under the conventional conditions. • formulate reasoned opinions in the mechanistic side of organic reactions.

Assessment and Evaluation: Attendance 10%, 3 tutorials 20% and year end final exam 70%.

Suggested Readings

1. A Guide Book to Mechanism in Organic Chemistry, P. Sykes, Orient Longman Ltd.
2. Advanced Organic Chemistry, J. March, McGraw Hill.
3. Organic Chemistry, I. L. Finar, Longman, Vol. 1 and 2, ELBS, Longman.
4. Stereochemistry of Carbon Compounds, E. Eliel, McGraw Hill.

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and Boyd, Prentice-Hall of India Pvt. Ltd.
Organic Chemistry, Fourth Ed. D. H. Williams and Ian Fleming, Tata McGraw-H
Compounds, P. S. Kalsi, Wiley Eastern Limited, India.
W. Kemp, ELBS with Macmillan.
Spectroscopy, M. Rabiul Islam and Mirza Aminul Huq, Ashrafia Boighar, Dhaka.
Spectroscopy, Mirza Aminul Huq and M. Rabiul Islam, Bangla Academy, Dhaka.

Code: Chem. 332F	**Course Title:** Chemistry of Natural Products	
	Credit: 4	**Unit:** 1
Marks: 10+20+70 = 100	**Contact Hours:** 55-60	**Session:** 2021-2022, 2022-2023
Type: Core Theory	**Course Teacher(s):** Will be assigned before course commencement.	

Rationale

This course is integrated of convey the fundamental concept of natural products, extraction, isolation and structu elucidation of different natural products and their properties.

Learning Objectives

Learning objectives of this course are to

- acquaint the students with the fundamental ideas of natural products chemistry.
- understand extraction, isolation and structural elucidation of terpenoids, carbohydrates, α-amino acids, puri and nucleic acids, alkaloids, steroids and antibiotics.
- explain the physical properties of terpenoids, carbohydrates, α-amino acids and proteins, purines and nucleic ac alkaloids, steroids and antibiotics.

Course Content	Learning Outcomes
1. Natural products: Definition, occurrence and importance of some natural products. 2. Terpenoids: Definition, classification; essential oils, isoprene rule, isolation and purification and general methods of determining structures of terpenoids. Detailed studies of (i) acyclic monoterpenoids like myrcene, citral; (ii) monocyclic monoterpenoids like limonene; (iii) sesquiterpenoids like farnesol. Biosynthesis of monoterpenoids. 3. Carbohydrates: Definition, classification; monosaccharides: structure, configuration, conformation and reactions of glucose; disaccharides: structures and conformations of sucrose and maltose; polysaccharides: preliminary ideas of starch and cellulose. 4. Amino acids and proteins: Structure, classification, synthesis, physical and chemical properties of α-amino acids; analysis of α-amino acids; peptides: structure and synthesis; proteins: general nature and structure (primary and secondary). 5. Purines and nucleic acids: Introduction; structure and synthesis of important derivatives of purine bases like adenine and guanine; structural elucidation of uric acid, nucleosides, nucleotides and nucleic acids; replication of DNA. 6. Alkaloids: Occurrence, classification, extraction and purification of alkaloids; general methods of determining structure; structural elucidation of ephedrine, atropine and morphine. 7. (a) Lipids: Definition, general classification, glycolipids, phospholipids, fats and oils, saponification number, iodine value; lipoproteins: definition, classification and their preliminary ideas; (b) Steroids: Definition, Diels' hydrocarbon, chemistry of cholesterol: functional group, angular methyl group and ring size determination. Preliminary ideas of steroidal glycosides: cardiotonic glycosides, saponins.	Upon completion of this cour students will be able to • describe the fundamentals of natural products chemistry. • explain extraction, isolation, structural elucidation of terpenoids, carbohydrates, α-amino acids, purines and nucleic acids, alkaloids, steroids and antibiotics. • discuss the chemistry of som important natural products lik terpenoids, carbohydrates, α-amino acids and proteins, purines and nucleic acids, alkaloids, steroids and antibiotics.

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8. Antibiotics: Definition, structural elucidation of penicillins and chloramphenicol by synthetic and degradative methods.	

Assessment and Evaluation: Attendance 10%, 3 tutorials 20% and year end final exam 70%.

Suggested Readings

1. Organic Chemistry by I. L. Finar, Longmans, Vol. 2, ELBS, Longman, Pearson.
2. Natural Product Chemistry, K. B. G. Torssell, J. Wiley and sons New York.
3. Selected Organic Synthesis, I. Fleming, J. Wiley and Sons, New York.
4. An Introduction to the Alkaloids, G. A. Wawan, Blackwell.
5. Mono and sesqui-terpenes, P. de Mayo, Inter Science.
6. Chemistry of the Alkaloids, S. W. Pelletier, von Nostrand Renihold.
7. Natural Products Chemistry, P. S. Kalsi, Kalyani Publishers, New Delhi.

Course Code: Chem. 334H	**Course Title:** Advanced Stereochemistry	
Year: 3rd	**Credit:** 2	**Unit:** 0.5
Marks: 5+10+35 = 50	**Contact Hours:** 35-40	**Session:** 2021-2022, 2022-2023
Type: Core Theory	**Course Teacher(s):** Will be assigned before course commencement.	

Rationale

This course in incorporated to provide advanced concepts of stereochemistry, stereochemistry of cyclic compounds, chirality of biomolecules, resolution of racemic mixture and application of CD, ORD, octant rule, ∝-haloketone rule.

Learning Objectives

Learning objectives of this course are to

- provide the fundamental concepts to the shapes of molecules, symmetry and streogenicity/ chairality, etc.
- extend the ideas to analyze the conformational properties of some important biopolymers.
- make them understand the basics of stereochemistry, physical properties of isomers and concepts of configuration and conformation.
- know assignment of (R) - and (S) - descriptions to streogenic centers in chiral molecules.
- provide in dept knowledge on conformational analyses of fused and bridged ring compounds, proteins, nucleic acid and carbohydrates.

Course Content	Learning Outcomes
1. *cis-trans* stereoisomerism in C=C and C=N compounds, interconversion of *cis-trans* isomers, physical properties; determination of configuration by physical and chemical methods; conformational analysis of acyclic compounds; configuration: relative and absolute configuration, determination of *R/S*, D/L configuration, chemical interconversion, chemical correlation methods, physical and spectral properties of diastereoisomers and conformers, conformation and reactivity. 2. Configuration and conformation of cyclic compounds; Three, four, five and six-membered ring conformational analysis and di- and poly-substituted cyclohexanes, inositols, conformation and physical properties, reactivity in cyclohexane and cyclohexanones derivatives, Winstein Eliel equation, Curtin-Hammet principle. 3. Conformational analysis of biopolymers: (a) Conformation of carbohydrates: anomeric effect, conformation of glycosides, starch, cellulose and effect of conformation on physical properties; (b) conformation of proteins and nucleic acids. 4. Chiral and prochiral molecules, pro-*R*, Pro-*S*, homotopic and heterotopic ligands and faces. Enantiotopic ligands (HCN addition), diastereotopic ligands and faces. enzymatic oxidation-reduction, prostereoisomerism in	Upon completion of this course students will be able to • describe the principle of stereochemistry. • interpret streochemically the pattern of reactivity on the basis of mechanistic reasoning. • demonstrate and analyze stereochemistry of some selected organic molecules in the biological system, viz carbohydrates, protein, and nucleic acids. • explain the conformation of fused and bridged ring compounds.

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biochemical reactions (citric acid cycle), stereochemistry of molecule synthetase reaction. 5. Chirality in molecules devoid of chiral centres: atropisomerism, biphenyls, allenes, cyclic allenes, molecules with planar chirality, cyclophanes, annulenes; chromatographic resolution, kinetic resolution and enzymatic resolution of racemic mixture. 6. Introduction and basic principles of optical rotatory dispersion curve, circular dichroisom, octant rule, α-haloketone rule and their application. 7. Fused rings and bridged rings: (a) Norboranes – bridged rings; (b) strained carbocycles-Bredt's rule; (c) fused rings: decalins, hydrindanes, steroid nucleus, bridged alkaloids.	

Assessment and Evaluation: Attendance 10%, 3 tutorials 20% and year end final exam 70%.

Suggested Readings

1. Stereochemistry of carbon compounds, El. Eliel, John Wiley and Sons.
2. Stereochemistry-Kagan, Edward Arnold.
3. Introduction to stereochemistry-Mislow, Benjamin.
4. Stereochemistry, conformation and Mechanism, P. S. Kalsi, Wiley Eastern Ltd.
5. Stereochemistry of Organic Compounds, Principles and Applications, D. Nasipuri, 4th ed. New Age Internatio Publications.
6. Organic chemistry, Morrison and Boyd, Pearson.
7. Organic Chemistry, I. L. Finar, Vol. 2, ELBS, Longman, Pearson.
8. Modern Stereochemistry (in Bangla), M R Islam, The Royal Publishers, Dhaka.

Course Code: Chem. 340F	**Course Title:** Chemical Spectroscopy	
Year: 3rd	**Credit:** 4	**Unit:** 1
Marks: 10+20+70 = 100	**Contact Hours:** 55-60	**Session:** 2021-2022, 2022-202
Type: Core Theory	**Course Teacher(s):** Will be assigned before course commencement.	

Rationale

This course is originated to impart concepts of the interaction between electromagnetic radiation and matter, principl various spectroscopic techniques, instrumentation and applications.

Course Objectives

Learning objectives of this course are to

- realize the interaction between electromagnetic radiation and matter.
- achieve knowledge on the fundamental principles of spectroscopic techniques.
- prepare students for advanced studies involving spectroscopic analysis of organic compounds.
- attain concepts of absorption, emission and chemical shift.
- understand basics of vibrational, rotational and other types of transitions.

Course Content	Learning Outcomes
1. Interaction of electromagnetic radiation with atoms and molecules: Electromagnetic radiation, quantization of energy, absorption and emission of radiation, regions of spectrum, representation of spectra, basic elements of practical spectroscopy, signal-to-noise: resolving power, line width, intensity of spectral transitions, removal of line broadening. 2. Microwave spectroscopy: Observational methods for the absorption of radiation by rotating molecules, rotation of molecules, rotational spectra; rotation of diatomic molecule, rotational spectra and bond length of diatomic molecule, rotation of polyatomic molecules, the Stark effect, techniques and instrumentation, microwave oven.	Upon completion of the cours students should be able to • know absorption and emis of radiation, region of spectrum and quantization energy. • calculate bond length of diatomic molecule, rotatio constant (B), ratio of population in different ene

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3. Vibrational spectroscopy: Vibrating diatomic molecule, diatomic vibrating rotator; vibration-rotation spectrum of carbon monoxide, breakdown of the Born-Oppenheimer approximation: interaction of rotations and vibrations; vibrations of polyatomic molecules. 4. Raman spectroscopy: Quantum theory of Raman effect, classical theory of Raman effect, pure rotational Raman spectra; vibrational Raman spectra; structure determination from Raman and infrared spectroscopy, techniques and instrumentation. 5. Electronic spectroscopy of atoms: Hydrogen atom spectrum, electronic angular momentum, fine structure of hydrogen atom spectrum; many-electron atoms: spectrum of lithium and other hydrogen-like species; angular momentum of many-electron atoms, term symbols, spectrum of helium and alkaline earths; atomic absorption spectroscopy. 6. Electronic spectroscopy of molecules: Electronic spectra of diatomic molecules; vibration coarse structure, Franck-Condon principle; dissociation energy and dissociation products; rotational fine structure of electronic – vibration transition, electronic angular momentum in diatomic molecules, spectrum of hydrogen molecule, chemical analysis by electronic spectroscopy; re-emission of energy by excited molecules: fluorescence and phosphorescence; circular dichroism (CD) spectroscopy. 7. Spin Resonance Spectroscopy: Spin and an applied magnetic field, interaction between spin and magnetic field, population of energy levels, lamor precession, relaxation times, sampling and instrumentation, ^{1}H nuclear magnetic resonance spectroscopy; chemical shift and factors affecting chemical shift, chemical analysis by NMR techniques, electron spin resonance spectroscopy: introduction, the position of ESR absorptions, the fine structure of ESR absorptions, hyperfine structure, double resonance in ESR, techniques of ESR spectroscopy; Mössbauer spectroscopy: instruction; techniques and applications.	states and resonance frequency. • understand the interaction of radiation with atoms/ molecules and between spin and magnetic field. • identify the molecules which will show microwave rotational spectrum, IR, NMR, ESR, CD and Mössbauer spectroscopy. • describe the allowed selection rule and transition between vibrational, rotational, electronic energy levels for atom/molecules. • state and explain different rules/ principles regarding chemical spectrocopy. • apply microwave, IR, Raman, electronic and spin resonance spectroscopy for chemical analysis.

Assessment and Evaluation: Attendance 10%, 3 tutorials 20% and year end final exam 70%.

Suggested Readings

1. Fundamentals of Molecular Spectroscopy, C. N. Banwell and E. M. McCash, 4th Edition.
2. Spectroscopy of Organic Chemistry, P. S. Kalsi, 2nd Edition.
3. Introduction to Molecular Spectroscopy, G. M. Barrow.
4. Modern Spectroscopy, J. Michael Hollas.
5. Spectroscopy (Atomic and Molecular), G. Chatwal and Shan Anand.
6. Molecular Spectroscopy, P. S. Sindhu.
7. Introduction to Spectroscopy, F. P. Larkins.
8. Basic Principles of Spectroscopy, R. Chang.
9. Introduction to Spectroscopy, D. L. Pavia, G. M. Lampman, G. S. Kriz

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Course Code: Chem. 315LF	Course Title: Physical Chemistry Practical III	
Year: 3rd	Credit: 4	Unit: 1
Marks: 10+20+70 = 100	Contact Hours: 55-60	Session: 2021-2022, 20
Type: Core Practical	Course Teacher(s): Will be assigned before course commencement.	

Rationale

This course provides practical knowledge on cryoscopy, partial molar volume, optical activity, rea spectrophotometric determination of concentration and conductometric titration.

Course Objectives

The learning objectives of this course are to:

- learn principle of different physico-chemical techniques and collect experimental data.
- organize collected experimental data.
- impart skill on various experimental techniques.
- analyse and interpret experimentally collected data.

Course Content	Learning Outcon
Expt. 1: To determine the molecular weight of a solute by cryoscopic method. Expt. 2: To study the adsorption of acetic acid on charcoal and examine the validity of Freundlich's adsorption isotherm and Langmuir's adsorption isotherm. Expt. 3: To study the liquid-liquid miscibility of three liquid system and to determine the position of one tie line. Expt. 4: Phase diagram of the two-component system: The pure components only crystallize from the solution. Expt. 5: To draw the phase diagram of the two-component system with continuous solid solution formation. Expt. 6: To determine the partial molal volume of solutions (Alcohol-Water system). Expt. 7: To find (a) the specific rotation of sucrose (b) concentration of an unknown sucrose solution (c) intrinsic rotation (d) influence of solvent on the rotation of sucrose solution by polarimetric method. Expt. 8: To determine the percentage of d-sucrose and d-tartaric acid in a given solutions polarimetrically. Expt. 9: To determine the specific reaction rate of the inversion of sucrose in presence of an acid by polarimetric method. Expt.10: To determine (a) the absorption curve of coloured substances in solution, (b) the validity of Beer-Lambert law, and (c) the strength of an unknown solution. Expt.11: To determine the concentration of potassium dichromate and potassium permanganate solution in a mixture by spectrophotometric method. Expt.12: To determine (a) the cell constant of a conductance cell, (b) the solubility of a sparingly soluble salt, and (c) the effect of salt on the solubility by conductometric method. Expt.13: To determine the specific reaction rate of base catalysed hydrolysis of an ester by conductometric method. Expt.14: To find out the equivalent conductance at infinite dilution and the dissociation constant of a weak electrolyte. Expt.15: Conductometric titration of (a) strong and (b) weak acids and to draw their neutralization curves. Expt.16: Potentiometric titration of (a) strong and (b) weak acids and to draw their neutralization curves. Expt.17: To determine the Fe^{2+} ion concentration in a given solution by potentiometric titration and to determine the value of E° for Fe^{2+}/Fe^{3+}. Expt.18: To determine the standard electrode potential of zinc and copper. N.B.: Experiments may be added to or omitted from the above list if necessary.	Upon completion of this the student will be able t • determine the molecu weight of a solid usin cryoscopic method. • determine the partial volume of ethanol. • determine the specifi of sucrose, reaction r hydrolysis of an este NaOH and E° for Fe^{2} • estimate the concentr $K_2Cr_2O_7$ and $KMnO_4$ solution mixture by spectrophotometric m • verify the validity o law and determine t strength of an unkno solution spectrophotometrica • calculate the solubilit sparingly soluble salt conductometrically. • draw a phase diagram partially miscible thr component system an neutralization curves conductometric titrati acid against base.

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Assessment and Evaluation: Attendance 10%, 3 tutorials 20% and year end final exam 70%.

Suggested Readings

1. Advanced Physical Chemistry Experiments, J. N. Gurtu and Amit Gurtu.
2. Practical Physical Chemistry, A. Findlay, Longmans.
3. Practical Physical Chemistry, Palit.
4. Practical Physical Chemistry, Sharma.
5. Advanced Practical Physical Chemistry, J. B. Yadav.

Course Code: Chem. 335LF	**Course Title:** Identification of Organic Compounds by Chemical and Spectroscopic Methods	
Year: 3rd	**Credit:** 4	**Unit:** 1
Marks: 10+20+70 = 100	**Contact Hours:** 55-60	**Session:** 2021-2022, 2022-2023
Type: Core Practical	**Course Teacher(s):** Will be assigned before course commencement.	

Rationale

This course is designed to convey knowledge on identification of organic compound by physical and chemical as well as spectroscopic methods.

Course Objectives

Learning objectives of this course are to

- identify organic compounds by physical and chemical methods.
- analyze spectral data to identify organic compounds using spectroscopic methods.

Course Content	Learning Outcomes
1. Physical appearance 2. Elemental analysis 3. Solubility Test 4. Functional group analysis by chemical and spectroscopic (IR, ^{1}HNMR) methods 5. Preparation of derivatives 6. Literature survey 7. Determination of Physical Constants (m.p, b.p. and mixed m.p determination). 8. Naming of the identified compounds and elucidating their structures by chemical as well as by spectroscopic methods.	Upon completion of this course students will be able to • find the elements in organic compounds • determine the functional groups. • analyze spectral data and finding from chemical and physical observation to elucidate structure of organic compounds.

Assessment and Evaluation: Attendance 10%, 3 tutorials 20% and year end final exam 70%.

Suggested Readings

1. Hand book of Organic Analysis, Qualitative and Quantitative, H. T. Clarke, B. Haynes, E. C. Brick, G. C. Shone, Edward Arnold, (5th Edition), Pub. Ltd. London.
2. Systematic Identification of Organic Compounds, R. L. Shriner, R. C. Fuson and D. Y. Curtin, John Wiley Sons, Inc. New York, London, Sydney, 5th Edition.
3. Text Book of Practical Organic Chemistry, Vogel's 5th Edition, ELBS with Longman.

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Course Code: Chem. 370LH	**Course Title:** Computing and information technology for chemistry	
Year: 3rd	**Credit:** 2	**Unit:** 0.5
Marks: 5+10+35 = 50	**Contact Hours:** 35-40	**Session:** 2021-2022, 2022
Type: Core Practical	**Course Teacher(s):** Will be assigned before course commencement.	

Rationale

This course is designed to convey the fundamental concepts of microsoft office, chem draw, and various p computer programmes to analyze and interpret data using computer software.

Course Objectives

The learning objectives of this course are to

- impart skill on various IT based computer programmes.
- analyze and interpret data using computer software.

Course Content	Learning Outcomes
Students will complete Part A, Part B and Part C of the course in 1st, 2nd and 3rd years respectively. Part A 1. a) Introduction to Windows and MS-DOS. b) Introduction to Microsoft Word. Writing chemistry texts involving chemical formulae, superscripts, subscripts, symbols. Part B 2. a) Advanced word processing-Tabs, tables, use of the equation editor in chemistry. b) Two-dimensional chemical structure drawing using Chem-Window. c) Microsoft Excel/Origin-Chemical spreadsheets (use of the function wizard, formulae) and graphs in chemistry (line graphs and scatter plots). Part C 3. a) Databases using Access. b) Use of PowerPoint. c) Molecular Structure Drawing using Chemwin 2D and 3D. d) Use of Internet. During the progress of the course, use of new software will be encouraged depending on availability.	Upon completion of this co the student will be able to • use MS word and MS exce origin, Chem Draw softwa • prepare salary sheet using a programme. • draw chemical structures. • prepare power-point presentation. • draw figure using excel and origin software.

Assessment and Evaluation: Attendance 10%, 3 tutorials 20% and year end final exam 70%.

Suggested Readings

1. Microsoft Office, 97/2000, Mahbubur Rahman, Cistech publications, 38/3 Bangla Bazar, Dhaka.
2. SAMS Teach Yourself Microsoft Office 2000 in 21 days, Laurie Ulrich, Techmedia, New Delhi-2.
3. Windows 98, Bruce A. Hallberg and Joe Casad, Techmedia, 20 Ansari Road, New Delhi-2.

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